

SAIKRISHNA PURANAM

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Education

The University of Texas at Dallas

May 2026

Bachelor of Science in Computer Science

GPA: 3.56

Relevant Coursework: Data Structures, OOP, Software Engineering, AI, ML, Probability & Statistics, Databases

Leadership: Residential Housing Association (RHA): **President**

Technical Skills

Programming Languages: Python, Java, C/C++, HTML/CSS/JavaScript, TypeScript, MATLAB

Frameworks: React, Next.js, Node.js, Express.js, Axios, Flask, PineCone, Scikit-Learn, PyTorch

Dev Tools: Docker, Kubernetes, Git, UNIX/Linux, VS Code, Jupyter Notebook, PyCharm, IntelliJ, Postman, Mocha, JUnit

Databases: MySQL, PostgreSQL, SupaBase, MongoDB

Certifications: AWS Cloud Practitioner

Experience

Algorithm Analysis Lead

September 2025 – Present

Artificial Muscle Prosthetic Research, UT Dallas

Richardson, TX

- Preprocessed multi-modal sensor data using **MATLAB**, generating samples for prosthetic arm movement control.
- Used signal processing with **PyWavelets** and **SciPy** to filter EMG data, reducing noise and improving input quality.
- Built **polynomial regression models** in scikit-learn and **statsmodels**, boosting joint movement prediction by 18%.
- Engineered features with scikit-learn and **tsfresh**, extracting 200+ biomechanical patterns to improve accuracy.
- Created validation pipelines with **GX Cloud** and tracked experiments in **MLflow**, lowering data errors by 10%.

Software Development Intern

May 2025 – August 2025

Mercatalyst

Carrollton, TX

- Automated Zendesk customer support responses using **Node.js** and **JavaScript**, reducing manual effort time by 60%.
- Refactored legacy code to utilize `async/await` from callbacks, resulting in a 40% improvement in system performance.
- Engineered REST API controllers for efficient CRUD operations and seamless **MongoDB** and **Salesforce** integration.
- Implemented multi-carrier ETA tracking in an open-source Node.js package, enhancing real-time accuracy and usability.
- Increased test coverage by 30% across microservices by through unit and integration tests using **Mocha** and **Puppeteer**, leveraging **Coveralls** and **ElasticSearch APM** for system wide monitoring.

Undergraduate Researcher

September 2024 – May 2025

"DDoS-as-a-Smokescreen" Research Lab, UT Dallas

Richardson, TX

- Utilized One-Class Encoder using **Scikit-Learn** for intrusion detection with 92% accuracy in identifying threats.
- Analyzed 20+GB of network traffic to classify events as benign or threats, enhancing system responsiveness.
- Developed anomaly detection algorithms using **Autoencoders** and **Isolation Forests**, reducing false positives.
- Optimized feature engineering for attack pattern recognition, improving detection rates by **12%**.

Software Development Intern

June 2024 - July 2024

Nagarro

Remote

- Led the development of a real-time analytics dashboard using **Next.js** and WebSockets within **Agile** sprints, integrating 20+ custom visualizations to deliver actionable insights for up to 1000 concurrent users with seamless performance.
- Engineered a backend API with **Java** and **Spring Boot**, designing scalable microservices for robust data aggregation.
- Containerized backend services with **Docker** and deployed via **Kubernetes**, ensuring 99.9% uptime during peak loads.
- Managed infrastructure with **AWS Elastic Beanstalk** and **EC2**, supporting feature releases on a weekly schedule.
- Improved code reliability by writing **JUnit** tests and resolving 12+ deployment bugs during QA and release cycles.

Projects

Crowd Capital | *Python, Flask, TensorFlow, ReactJS, Axios*

- Engineered a banking application for people living in banking deserts using **Python** and **ReactJS** in HackUTD.
- Developed a fraud detection system with **Gemini AI** and **Pinecone** for similarity searches, achieving 90% accuracy.
- Designed a community loan platform with **SupaBase**, enabling faster loan approvals via user contributions and voting.
- Built a credit score predictor using **RandomForestRegressor** and **TensorFlow**, achieving 85% accuracy
- Leveraged **Chart.js** and **D3.js** for dynamic visualizations, integrated via RESTful APIs using **Axios**.